

Things to Know

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Here are some things to know when reading about the Simple Unified Model.

Algebra of Sets

An *algebra of sets* on a set S is a collection of subsets of S that is closed under union and complement. We assume the empty set belongs to any algebra.

Exercise. *Show algebras of sets are closed under intersection.*

Hint. Use De Morgan's law.

Partition

If the algebra $\mathcal{A}$ is finite then the atoms of the algebra form a *partition* of S . A set $A \in \mathcal{A}$ is an *atom* of $\mathcal{A}$ if $B \subseteq A$ and $B \in \mathcal{A}$ imply $B = \emptyset$ or $B = A$.

Exercise. *Show $A_s = \cap\{A \in \mathcal{A} \mid s \in A\}$ is an atom of $\mathcal{A}$.*

For $s \in S$ the *atom* containing s is the intersection of all sets in the algebra containing s , $A_s = \cap\{A \in \mathcal{A} \mid s \in A\}$.

Exercise. *Show if $B \subseteq A_s$ and $B \in \mathcal{A}$ then either $B = \emptyset$ or $B = A_s$.*

Hint: If $s \in B$ then $A_s \subseteq B$. If $s \notin B$ then $s \in A_s \setminus B$ so $A_s \subseteq A_s \setminus B \subseteq A_s$. It follows $A_s = A_s \setminus B$ and $B = \emptyset$.

Clearly $S = \cup_{s \in S} A_s$.

Exercise. *Show $A_s \cap A_t \neq \emptyset$ implies $A_s = A_t$.*

Hint: Use $A_s \subseteq A_t$ and $A_t \subseteq A_s$ then apply the previous exercise.

Partial Information

A partition models *partial information* about a sample space S . No information corresponds to the singleton $\{S\}$. Full information corresponds to the partition of singletons $\{\{s\} \mid s \in S\}$. Partial information means knowing only which atom contains the element s .

For example, the partition $\{[0, 1/2), [1/2, 1)\}$ of $[0, 1)$ represents knowledge of the first digit in the base 2 expansion of $s \in [0, 1)$. If $s \in [0, 1/2)$ its first binary digit is 0 and if $s \in [1/2, 1)$ its first binary digit is 1.

Measurable

A function $f: S \rightarrow \mathbf{R}$ is *measurable* with respect to an algebra if and only if it is constant on atoms. In particular, it is a function on the atoms. This is useful for software implementation of measurable functions.

Finitely Additive Measure

The set of bounded functions on a set S is $B(S) = \{f: S \rightarrow \mathbf{R} \mid \|f\| < \infty\}$ where $\|f\| = \sup_{s \in S} |f(s)|$ is the norm of f .

The vector space dual of bounded functions, $B(S)^*$, is isometrically isomorphic to $ba(S)$, the space of finitely additive measures on S . Given a linear functional $L: B(S) \rightarrow \mathbf{R}$ define $\lambda(A) = L1_A$. The algebraic equality $1_{A \cup B} = 1_A + 1_B - 1_{A \cap B}$ shows λ is a finitely additive measure.

Exercise. Show $\lambda(\emptyset) = 0$.

Hint: $1_\emptyset(s) = 0$ for all $s \in S$ so $1_\emptyset$ is the zero function in $B(S)$.

Exercise. If $T: V \rightarrow W$ is a linear operator then $T0_V = 0_W$.

Hint First show $v + v = v$ for any $v \in V$ implies $v = 0_v$. Consider $T(0_V) = T(0_V + 0_v)$.

Given a finitely additive measure $\lambda \in ba(S)$ we can define a linear functional on the vector space of finite linear combinations $f = \sum_j a_j 1_{A_j}$ by $Lf = \sum_j a_j \lambda(A_j)$. The A_j can be chosen to be disjoint.

Exercise. Write $A \cup B$ as a disjoint union.

Hint: Use $A \setminus B = \{a \in A \mid a \notin B\}$.

This makes the definition well-defined. The norm of λ is the supremum over all disjoint $\{A_j\}$ of $\sum_j |\lambda(A_j)|$. A short argument shows $\|\lambda\| = \|L\|$.

Elementary functions are uniformly dense in $B(S)$ so we can extend L to an isometry from $B(S)$ to $ba(S)$.

For $g \in B(S)$ define $M_g: B(S) \rightarrow B(S)$ by $M_g f = fg$. Its adjoint $M_g^*: ba(S) \rightarrow ba(S)$ defines multiplication of a measure by a bounded function $g\lambda = M_g^* \lambda$.

If P is a positive measure having mass 1 (a “probability” measure) and $\mathcal{A}$ is an algebra of sets on the sample space then the *conditional expectation* $Y = E[X | \mathcal{A}]$ is equivalent to $Y(P|_{\mathcal{A}}) = (XP)|_{\mathcal{A}}$ where $|_{\mathcal{A}}$ indicates restriction of measure. It is trivial to implement restriction of a function in software.

Fréchet Derivative

If $F: X \rightarrow Y$ is a function between normed linear spaces the Fréchet derivative $DF: X \rightarrow \mathcal{B}(X, Y)$ is defined by

$$F(x+h) = F(x) + DF(x)h + o(h)$$

for $x, h \in X$ where $\mathcal{B}(X, Y)$ is the set of bounded linear operators from X to Y . The “little o ” notation means $\|F(x+h) - F(x) - DF(x)h\|/\|h\| \rightarrow 0$ as $\|h\| \rightarrow 0$. The function F can be approximated near x by the linear operator $DF(x)$.

Exercise. If X is a Banach algebra and $F(x) = x^2$ show $DF(x)h = xh + hx$.

Hint: Don’t assume multiplication is commutative and use $h^2 = o(h)$.

If we define left multiplication by $x \in X$, $L_x: X \rightarrow X$, as $L_x y = xy$ and right multiplication by x as $R_x y = yx$ then the above exercise shows $DF(x^2) = L_x + R_x$.

Exercise Show $D(x^n) = \sum_{k=0}^{n-1} L_x^{n-k} R_x^k$.

For $x \in \mathbf{R}^n$ define the linear functional $x^*: \mathbf{R}^n \rightarrow \mathbf{R}$ by $x^*(y) = y \cdot x$.

Exercise. Show $x \mapsto x^*$ is an isometric isomorphism from $\mathbf{R}^n$ to its dual $(\mathbf{R}^n)^* = \mathcal{B}(\mathbf{R}^n, \mathbf{R})$.

Exercise Let $\|x\|^p = \sum_i |x_i|^p$ for $x = (x_1, \dots, x_n) \in \mathbf{R}^n$. Show $D\|x\|^p = p\|x\|^{p-2}x^*$.

Hint: Start with $p = 2$ and use $\|x\|^p = \exp((p/2) \log \|x\|^2)$. Note $x \mapsto \|x\|^p$ is a function $\mathbf{R}^n \rightarrow \mathbf{R}$ so its Fréchet derivative goes from $\mathbf{R}^n \rightarrow \mathcal{B}(\mathbf{R}^n, \mathbf{R}) = (\mathbf{R}^n)^*$.